Metadata and Docs URLs

You can customize several metadata configurations in your **FastAPI** application.

Metadata for API

You can set the following fields that are used in the OpenAPI specification and the automatic API docs UIs:

Parameter	Type	Description
`title`	`str`	The title of the API.
`summary`	`str`	A short summary of the API. Available since OpenAPI `3.1.0`, FastAPI `0.99.0`.
`description`	`str`	A short description of the API. It can use Markdown.
`version`	`string`	The version of the API. This is the version of your own application, not of OpenAPI. For example `2.5.0`.
`terms_of_service`	`str`	A URL to the Terms of Service for the API. If provided, this has to be a URL.
`contact`	`dict`	The contact information for the exposed API. It can contain several fields.

`contact` fields

Parameter	Type	Description
`name`	`str`	The identifying name of the contact person/organization.
`url`	`str`	The URL pointing to the contact information. MUST be in the format of a URL.
`email`	`str`	The email address of the contact person/organization. MUST be in the format of an email address.

`license_info`	`dict`	The license information for the exposed API. It can contain several fields.

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Parameter	Type	Description
		`license_info` **fields**

Parameter	Type	Description
`name`	`str`	**REQUIRED** (if a `license_info` is set). The license name used for the API.
`identifier`	`str`	An SPDX license expression for the API. The `identifier` field is mutually exclusive of the `url` field. Available since OpenAPI `3.1.0`, FastAPI `0.99.0`.
`url`	`str`	A URL to the license used for the API. MUST be in the format of a URL.

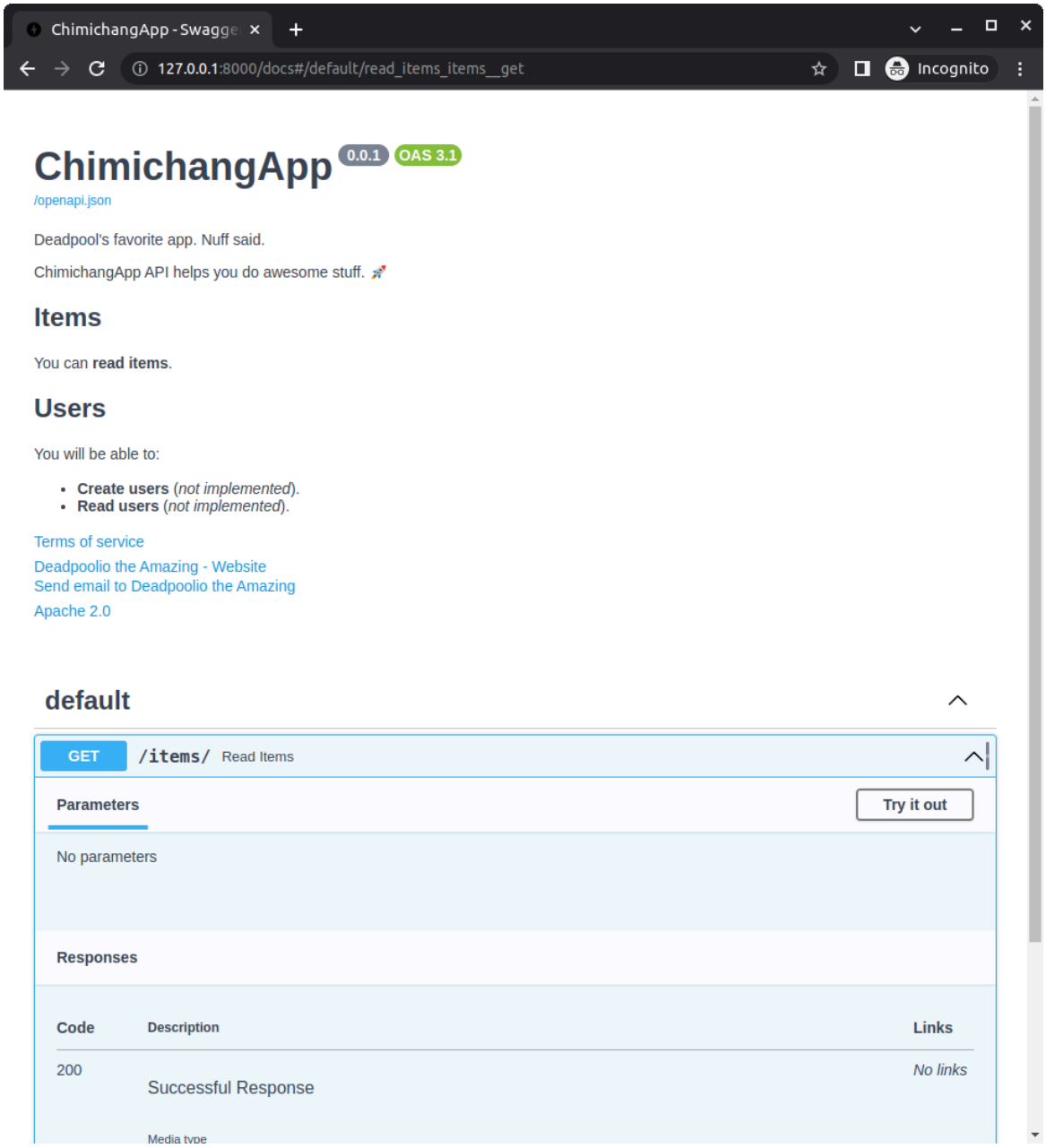
You can set them as follows:

Python 3.10+

```python
from fastapi import FastAPI

description = """
ChimichangApp API helps you do awesome stuff. 🚀

## Items

You can **read items**.

## Users

You will be able to:

* **Create users** (_not implemented_).
* **Read users** (_not implemented_).
"""

app = FastAPI(
    title="ChimichangApp",
    description=description,
    summary="Deadpool's favorite app. Nuff said.",
    version="0.0.1",
    terms_of_service="http://example.com/terms/",
    contact={
        "name": "Deadpoolio the Amazing",
        "url": "http://x-force.example.com/contact/",
        "email": "dp@x-force.example.com",
    },
    license_info={
        "name": "Apache 2.0",
        "url": "https://www.apache.org/licenses/LICENSE-2.0.html",
    },
)


@app.get("/items/")
async def read_items():
    return [{"name": "Katana"}]
```

> **Tip**
>
> You can write Markdown in the `description` field and it will be rendered in the output.

With this configuration, the automatic API docs would look like:



License identifier

Since OpenAPI `3.1.0` and FastAPI `0.99.0`, you can also set the `license_info` with an `identifier` instead of a `url`.

For example:

Python `3.10+`

```python
from fastapi import FastAPI

description = """
ChimichangApp API helps you do awesome stuff. 🚀

## Items

You can **read items**.

## Users

You will be able to:

* **Create users** (_not implemented_).
* **Read users** (_not implemented_).
"""

app = FastAPI(
    title="ChimichangApp",
    description=description,
    summary="Deadpool's favorite app. Nuff said.",
    version="0.0.1",
    terms_of_service="http://example.com/terms/",
    contact={
        "name": "Deadpoolio the Amazing",
        "url": "http://x-force.example.com/contact/",
        "email": "dp@x-force.example.com",
    },
    license_info={
        "name": "Apache 2.0",
        "identifier": "Apache-2.0",
    },
)


@app.get("/items/")
async def read_items():
    return [{"name": "Katana"}]
```

Metadata for tags

You can also add additional metadata for the different tags used to group your path operations with the parameter `openapi_tags`.

It takes a list containing one dictionary for each tag.

Each dictionary can contain:

- `name` (**required**): a `str` with the same tag name you use in the `tags` parameter in your *path operations* and `APIRouter`s.
- `description`: a `str` with a short description for the tag. It can have Markdown and will be shown in the docs UI.
- `externalDocs`: a `dict` describing external documentation with:
  - `description`: a `str` with a short description for the external docs.
  - `url` (**required**): a `str` with the URL for the external documentation.

Create metadata for tags

Let's try that in an example with tags for `users` and `items`.

Create metadata for your tags and pass it to the `openapi_tags` parameter:

Python 3.10+

```python
from fastapi import FastAPI

tags_metadata = [
    {
        "name": "users",
        "description": "Operations with users. The **login** logic is also here.",
    },
    {
        "name": "items",
        "description": "Manage items. So _fancy_ they have their own docs.",
        "externalDocs": {
            "description": "Items external docs",
            "url": "https://fastapi.tiangolo.com/",
        },
    },
]

app = FastAPI(openapi_tags=tags_metadata)


@app.get("/users/", tags=["users"])
async def get_users():
    return [{"name": "Harry"}, {"name": "Ron"}]


@app.get("/items/", tags=["items"])
async def get_items():
    return [{"name": "wand"}, {"name": "flying broom"}]
```

Notice that you can use Markdown inside of the descriptions, for example "login" will be shown in bold (**login**) and "fancy" will be shown in italics (*fancy*).

Tip

You don't have to add metadata for all the tags that you use.

Use your tags

Use the `tags` parameter with your *path operations* (and `APIRouter`s) to assign them to different tags:

Python 3.10+

```python
from fastapi import FastAPI

tags_metadata = [
    {
        "name": "users",
        "description": "Operations with users. The **login** logic is also here.",
    },
    {
        "name": "items",
        "description": "Manage items. So _fancy_ they have their own docs.",
        "externalDocs": {
            "description": "Items external docs",
            "url": "https://fastapi.tiangolo.com/",
        },
    },
]

app = FastAPI(openapi_tags=tags_metadata)


@app.get("/users/", tags=["users"])
```

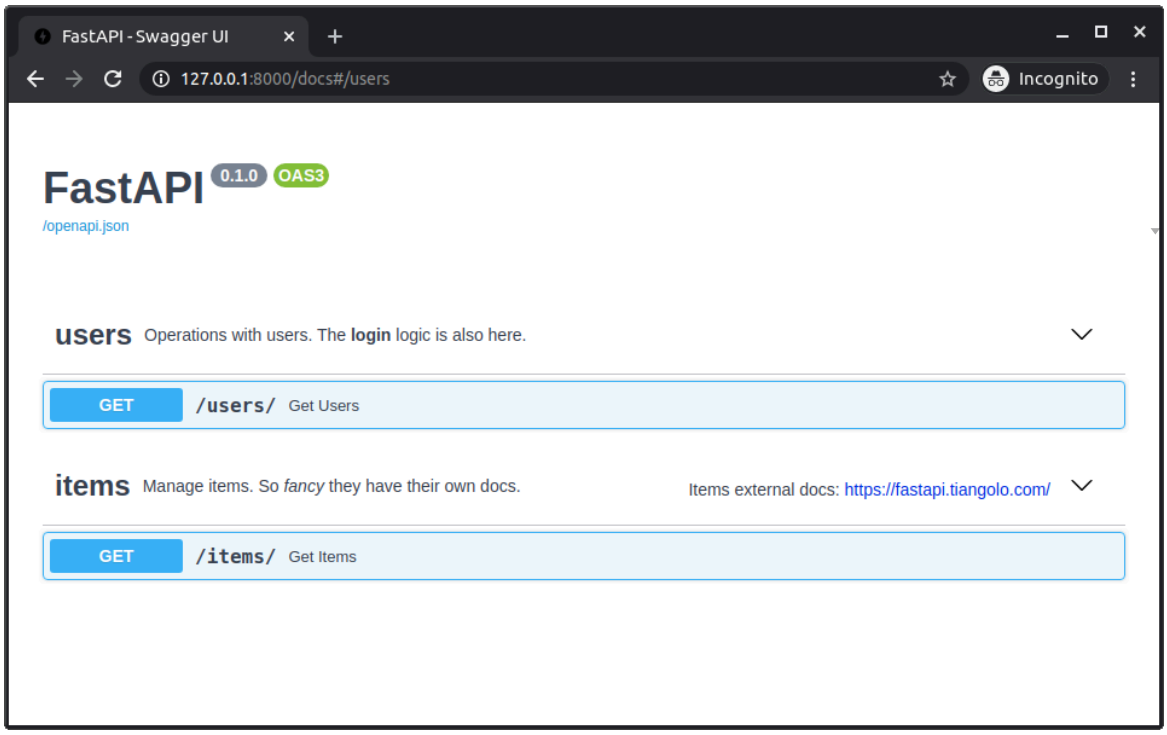
```python
async def get_users():
    return [{"name": "Harry"}, {"name": "Ron"}]


@app.get("/items/", tags=["items"])
async def get_items():
    return [{"name": "wand"}, {"name": "flying broom"}]
```

Info

Read more about tags in Path Operation Configuration.

Check the docs

Now, if you check the docs, they will show all the additional metadata:



Order of tags

The order of each tag metadata dictionary also defines the order shown in the docs UI.

For example, even though `users` would go after `items` in alphabetical order, it is shown before them, because we added their metadata as the first dictionary in the list.

OpenAPI URL

By default, the OpenAPI schema is served at `/openapi.json`.

But you can configure it with the parameter `openapi_url`.

For example, to set it to be served at `/api/v1/openapi.json`:

Python 3.10+

```python
from fastapi import FastAPI

app = FastAPI(openapi_url="/api/v1/openapi.json")


@app.get("/items/")
async def read_items():
    return [{"name": "Foo"}]
```

If you want to disable the OpenAPI schema completely you can set `openapi_url=None`, that will also disable the documentation user interfaces that use it.

Docs URLs

You can configure the two documentation user interfaces included:

- **Swagger UI**: served at `/docs`.
  - You can set its URL with the parameter `docs_url`.
  - You can disable it by setting `docs_url=None`.
- **ReDoc**: served at `/redoc`.
  - You can set its URL with the parameter `redoc_url`.
  - You can disable it by setting `redoc_url=None`.

For example, to set Swagger UI to be served at `/documentation` and disable ReDoc:

Python 3.10+

```python
from fastapi import FastAPI

app = FastAPI(docs_url="/documentation", redoc_url=None)


@app.get("/items/")
async def read_items():
    return [{"name": "Foo"}]
```